TEMASEK JUNIOR COLLEGE
Preliminary Examinations
Higher 2

BIOLOGY 9648/01

Paper 1 Multiple Choice

21st September 2011

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

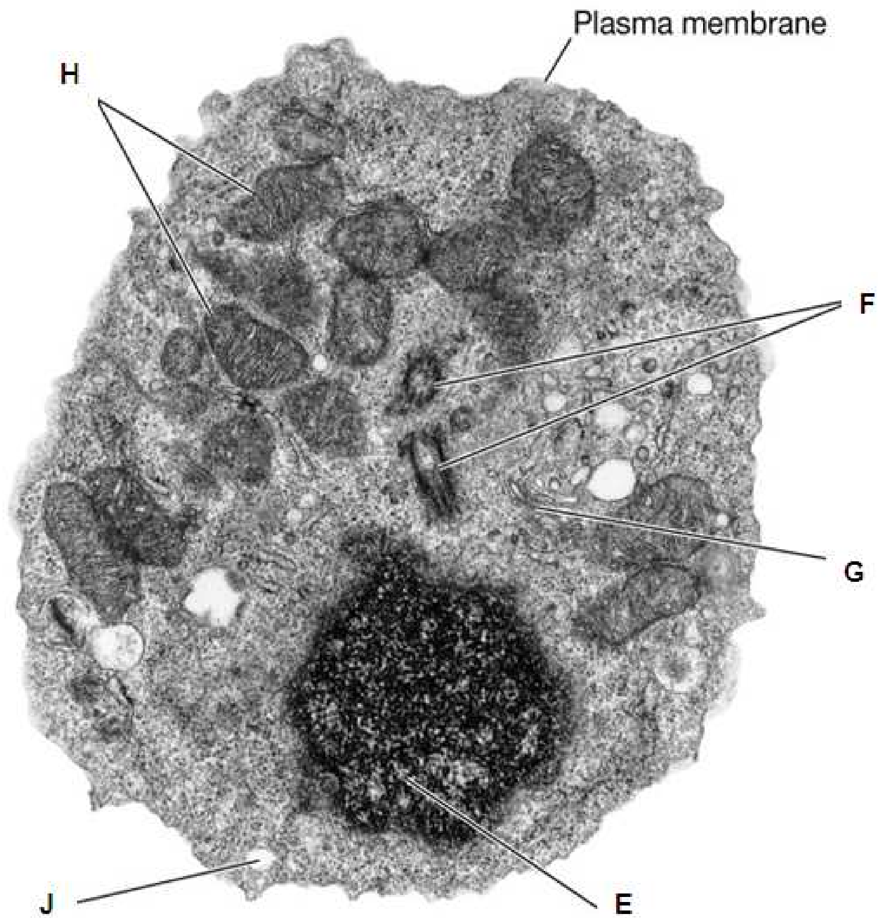
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Calculators may be used.

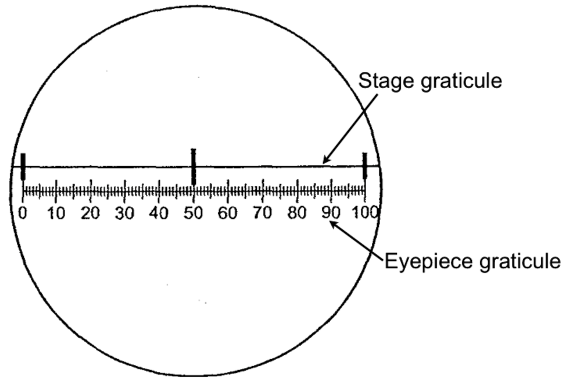
1 An electron micrograph of a cell is shown below.

Match the organelles E, F, G, H and J associated with the cellular processes listed.



	E	F	G	H	J
A	DNA replication	Digestion of material	Organizes the spindle	Oxidative phosphorylation	Packaging of secretory products
B	Oxidative phosphorylation	Organizes the spindle	Digestion of material	DNA replication	Packaging of secretory products
C	Organizes the spindle	Digestion of material	Oxidative phosphorylation	Packaging of secretory products	DNA replication
D	DNA replication	Organizes the spindle	Packaging of secretory products	Oxidative phosphorylation	Digestion of material

2 The figure below shows a stage graticule viewed through a light microscope fitted with an eyepiece graticule. The stage graticule has divisions of 0.1 mm.



The same microscope is then used to estimate the diameter of a plant cell, which measures approximately 67 smallest divisions on the eyepiece graticule.

What is the approximate diameter of the plant cell?

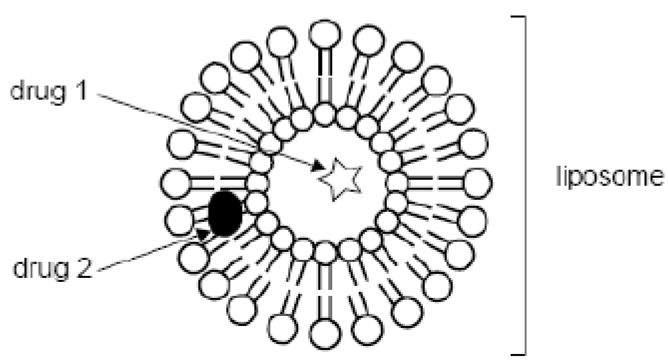
A 0.13 µm

B 1.3 µm

C 13 µm

D 130 µm

3 Liposomes can be found in the cytosol and are formed from a double layer of phospholipid molecules identical to those found in plasma membranes. Liposomes can also be manufactured and used to carry medicinal drugs into cells.



By examining the figure above, it is reasonable to expect that

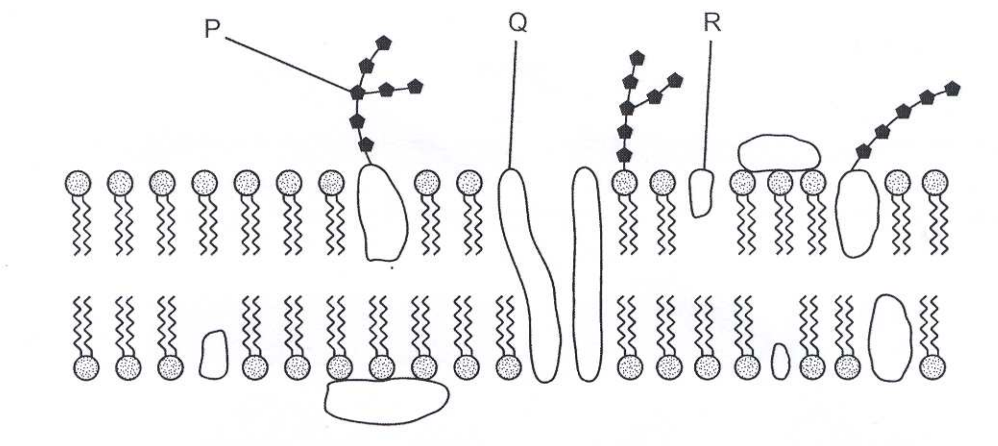
A drug **1** is lipophilic.

B drug **2** is water soluble.

C the interior of the liposome is aqueous.

D liposome-based membranes have a rigid structure.

4 The diagram below shows part of a cell surface membrane.



What is the correct function for each of the structures labelled?

	Regulates membrane fluidity	Forms hydrogen bonds with water to stabilise membrane	Transports ions and large polar molecules
A	P	R	Q
B	P	Q	R
C	Q	R	P
D	R	P	Q

5 Which of the following reactions illustrates hydrolysis?

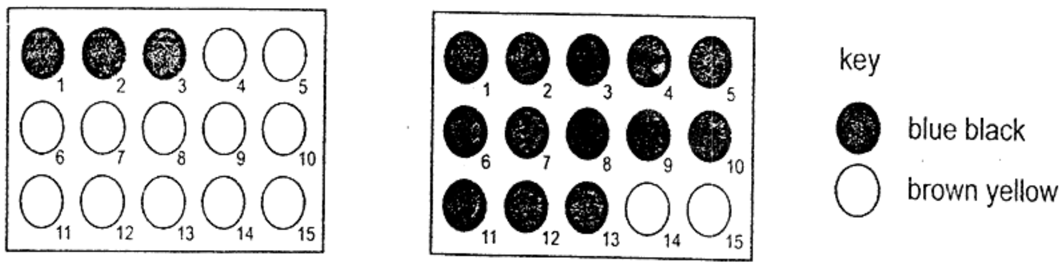
A The reaction of two monosaccharides to form a disaccharide with the release of water.

B The breakdown of a nucleotide to form a phosphate, a ribose sugar and a nitrogenous base with the production of a molecule of water.

C The synthesis of two amino acids to form a dipeptide with production of one molecule of water.

D The breakdown of a triglyceride molecule to give glycerol and fatty acids with the utilization of three molecules of water.

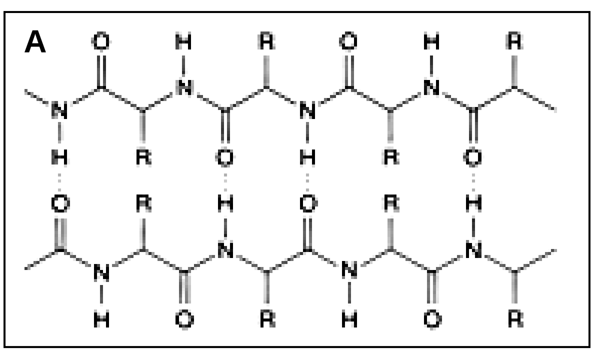
6 An experiment was carried out to investigate the digestion of starch using amylase at two different temperatures. A sample was removed from each mixture at 15 second intervals and placed onto a spotting tile well containing two drops of iodine in KI solution. The results are shown in the diagram.

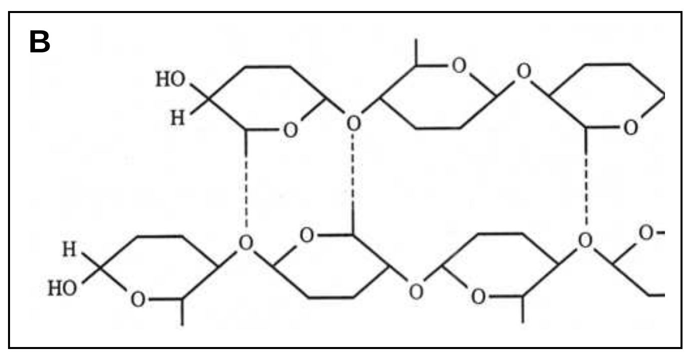


Which of the following shows the correct temperatures and times for the complete digestion of starch?

	temperature/ ^{o}C	time/s
A	30	3.15
	10	0.45
B	30	45
	10	195
C	10	45
	30	195
D	10	3.15
	30	0.45

7 The following shows two molecules that perform structural functions.





Which of the following statements does not describe a similarity between molecules A and B?

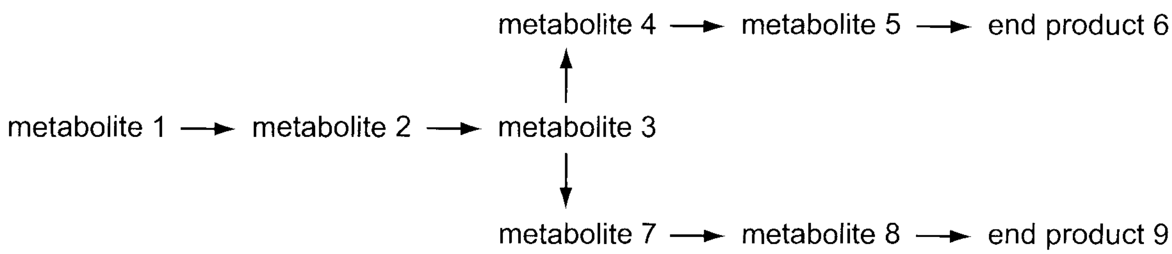
A Hydrogen bonds form cross-linkages between neighbouring chains.

B Monomers are joined together via peptide bonds.

C Monomers can be extracted by hydrolysing the bonds within each chain.

D Covalent bonds are present in both molecules.

8 The diagram shows a metabolic pathway controlled by end product inhibition. As the concentration of the end product increases above a set level, an enzyme in the pathway leading to the end product is inhibited.



An increase in the concentration of end product 6 does not lead to a decrease in the synthesis of end product 9.

Which enzyme is inhibited by end product 6?

A metabolite 1 → metabolite 2

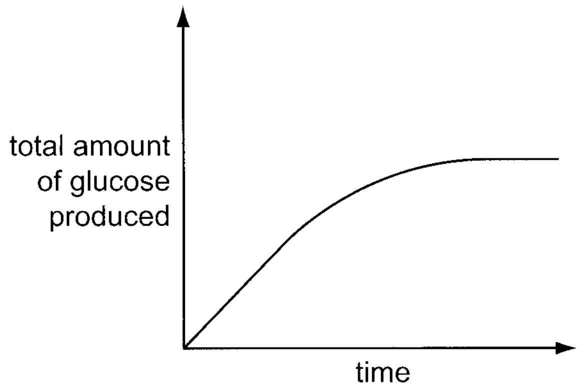
B metabolite 2 → metabolite 3

C metabolite 3 → metabolite 4

D metabolite 3 → metabolite 7

9 Lactose is a disaccharide present in milk. The enzyme β galactosidase catalyses the conversion of lactose to glucose and galactose.

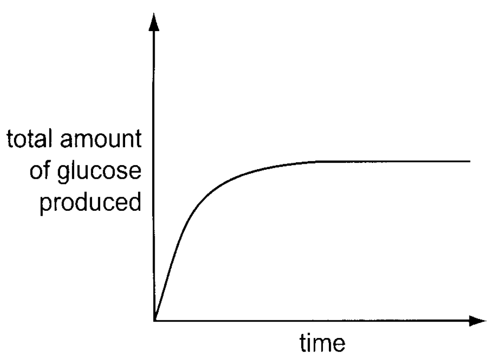
10 cm^3 of a 1% β galactosidase solution was added to 10 cm^3 of milk. The graph shows the total amount of glucose produced over the next ten minutes.



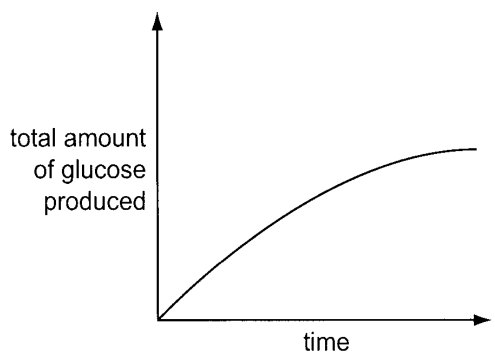
In another experiment, 10 cm^3 of a 2% β galactosidase solution was added to 10 cm^3 of milk.

Which graph shows the results that would be obtained?

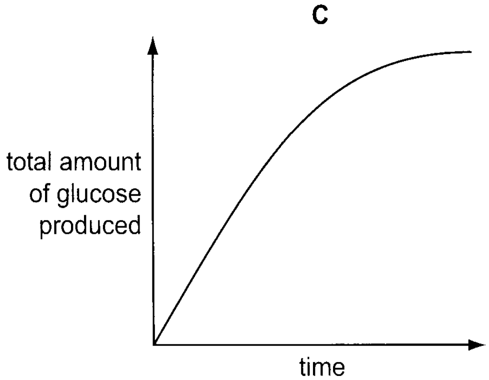
A



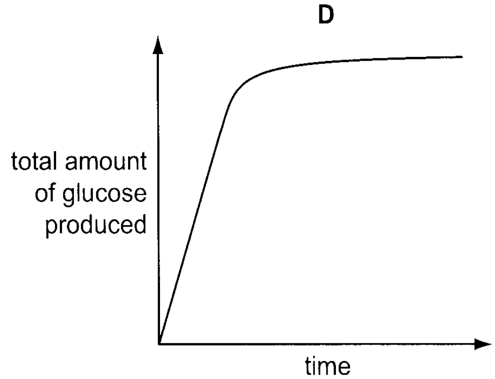
B



C

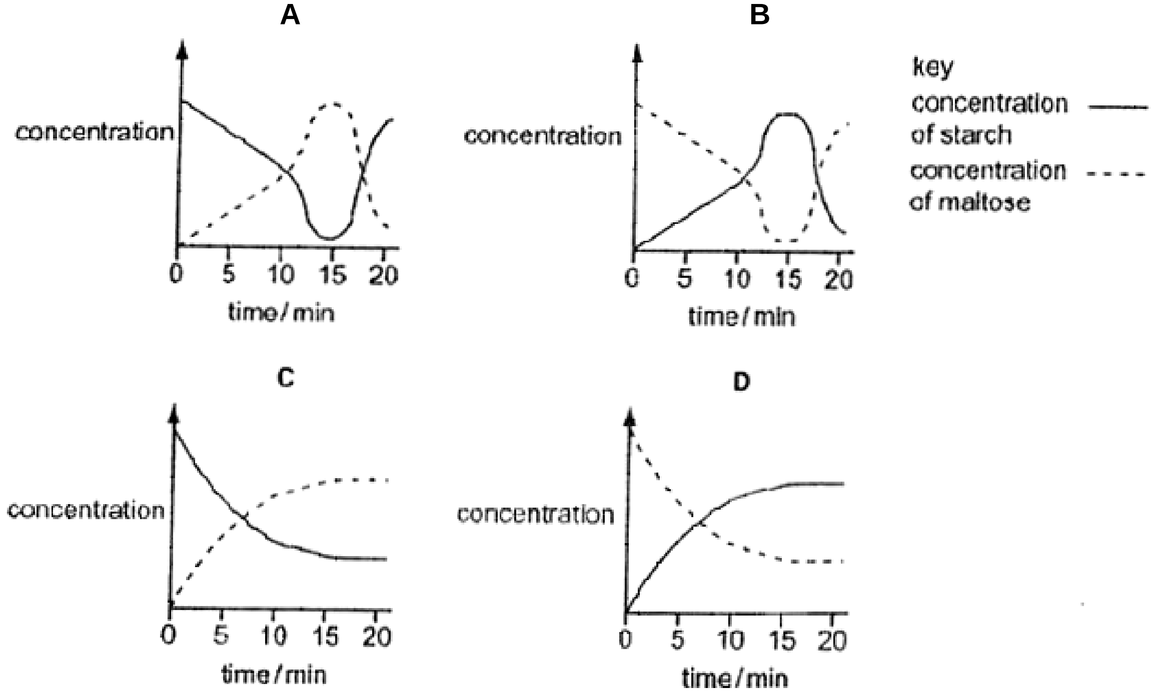


D

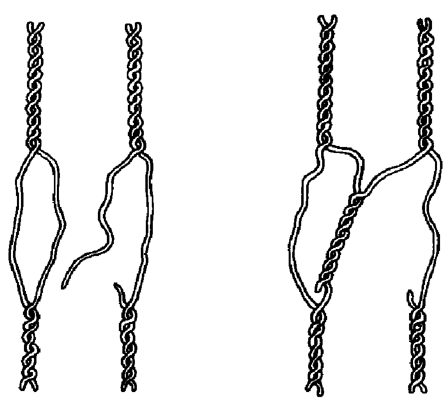


10 In an experiment using starch and amylase, the concentrations of starch and maltose present in the reacting mixture are measured every minute for 20 minutes. 1% hydrochloric acid is added after 10 minutes and the mixture is heated to 60°C at 14 minutes.

Which graph represents the results of this experiment?



11 The diagram shows the uncoiling and coiling that might take place between homologous regions of two double helical nuclei acid molecules.



What does the diagram represent?

A DNA replication

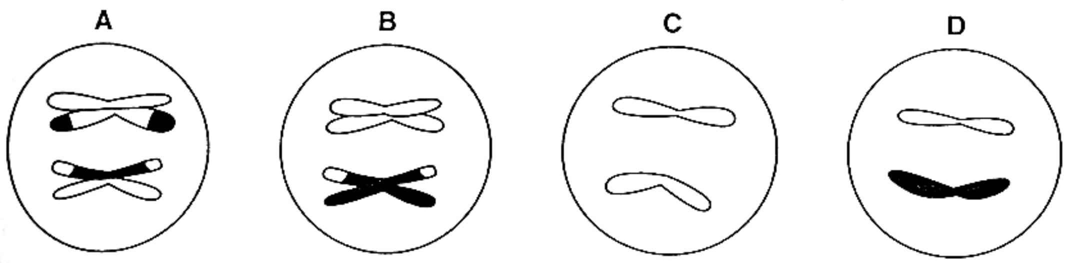
B mRNA synthesis

C Chromosome mutation

D The early stages of crossing over

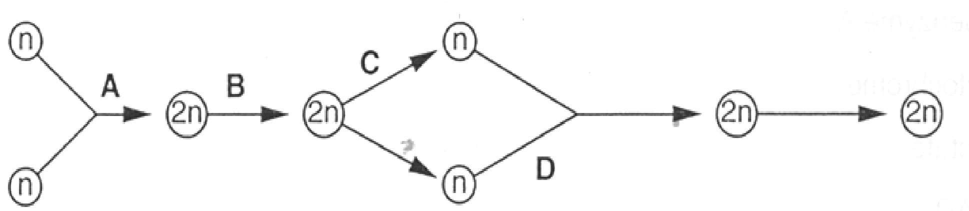
12 The diagram shows a cell at anaphase 1 of meiosis.

Which diagram shows a normal gamete that could be produced from this cell?

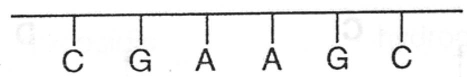


13 The diagram represents the life cycle of an animal.

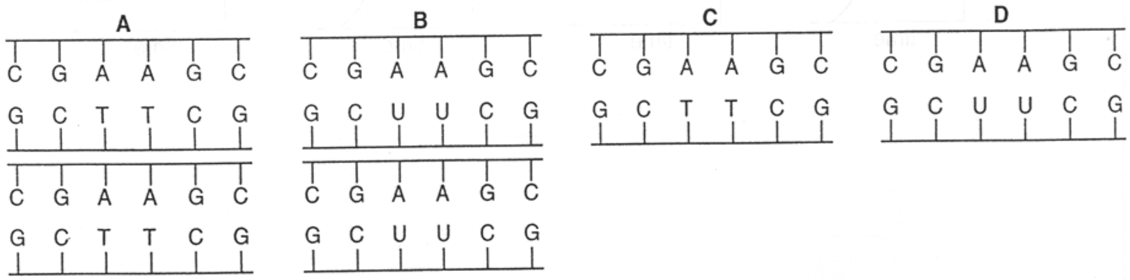
At which stage in the life cycle does mitosis occur?



14 The diagram represents one strand of part of a DNA molecule.



Which of the following would this section and its complementary strand give rise to after semi-conservative replication?



15 A gene has the following sequence on the template strand:

1 5 10 15

3' – C T G G C A T G C C T T C G G A A A -5'

Which of the following would probably result in the least change in the activity of the protein coded by this gene?

A Substitution of A for G in position 3

B Deletion of C at position 5

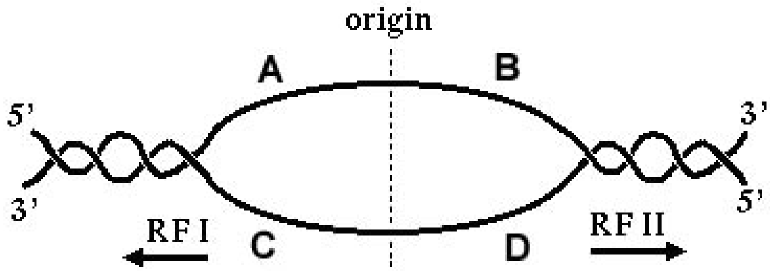
C Deletion of A at position 16

D Addition of G between positions 14 and 15

16 A new life form discovered on Mars has a genetic code similar to those on Earth, except there are five different DNA bases and the sequences are translated as doublets instead of triplets. How many different codons can be specified by this genetic code?

A 5 **B** 25 **C** 32 **D** 64

17 In the figure below, which DNA segment would serve as a template for continuous synthesis with DNA polymerase proceeding in the direction of replication fork I (RFI)?



18 The following are characteristics of eukaryotic transcription.

- Promoters are activated by transcription factors that recognize specific DNA sequences and other sequences that are very similar.
- Within a promoter, there may be recognition sites for more than one transcription factor.
- Similar specific DNA sequences can be recognized by more than one transcription factor.
- Each transcription factor may be capable of recognizing a number of promoter recognition sites.

What explains the different levels of expression of a eukaryotic gene?

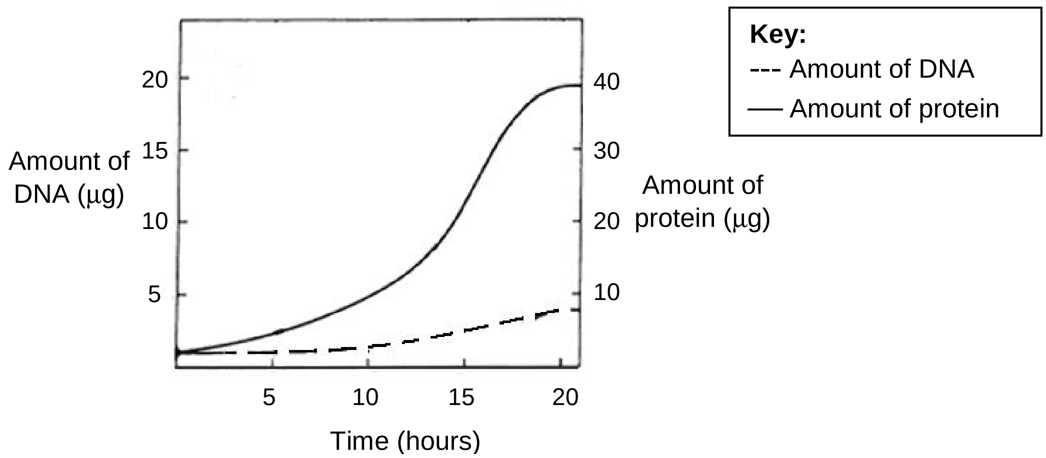
A Competition between recognition sites present in the promoter for transcription factors

B Competition between transcription factors that recognise the same sites of a promoter

C The number of transcription factors that recognize the same sites of a promoter

D The number of types of different transcription factors

19 During egg production in *Drosophila*, there is an increased production of eggshell proteins. The genes which code for the eggshell proteins are clustered on the X chromosome.



Which of the following best explains the trends observed in the graph?

A Gene amplification has occurred, increasing the rate of transcription. Cytoplasmic poly(A) tail addition has occurred, increasing the duration of translation.

B DNA replication has occurred via the rolling circle mechanism, increasing the rate of and duration of translation.

C Gene amplification has occurred, increasing the rate of mutation. Cytoplasmic poly(A) tail addition has occurred, increasing the duration of translation.

D DNA replication has occurred. The abundance and activity of eukaryotic translation initiation factors are increased simultaneously, thereby increasing the rate of translation.

20 What are the correct characteristics for a prokaryotic genome?

	Promoters	DNA always bound to histone proteins	Plasmids often present	Repeat sequences absent or uncommon
A	✓	✗	✗	✗
B	✗	✓	✓	✓
C	✗	✓	✗	✗
D	✓	✗	✓	✓

21 Cancer cells divide out of control, forming tumours.

Which statement describes the difference between a cancer cell and a normal cell?

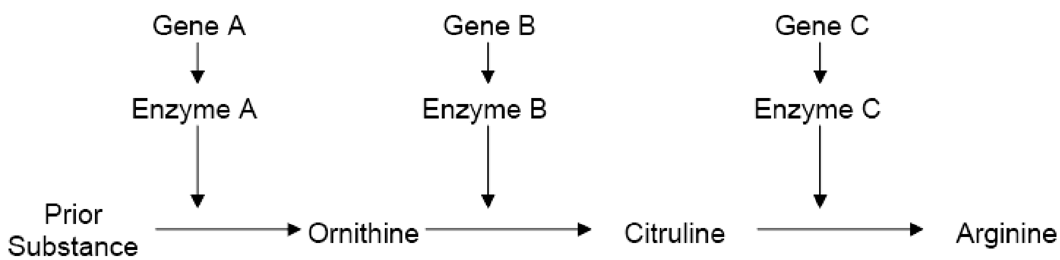
A Cancer cells do not undergo cytokinesis

B Cancer cells have a shorter interphase

C Cancer cells do not have metaphase

D Only cancer cells have mutated DNA

22 The bread mould, *Neurospora*, normally produces its own amino acids (ornithine, citruline and arginine) from raw materials through a system of enzymes.



If a nutritional mutant of *Neurospora* can only survive when provided with citruline, the mutation probably affected the following gene which could

A only be gene **A**

B only be gene **B**

C only be gene **C**

D be either gene **A** or **B**

23 In generalised transduction, defective virus are formed as a result of

A viral enzymes cutting the host DNA such that the host DNA is assembled into the new virus.

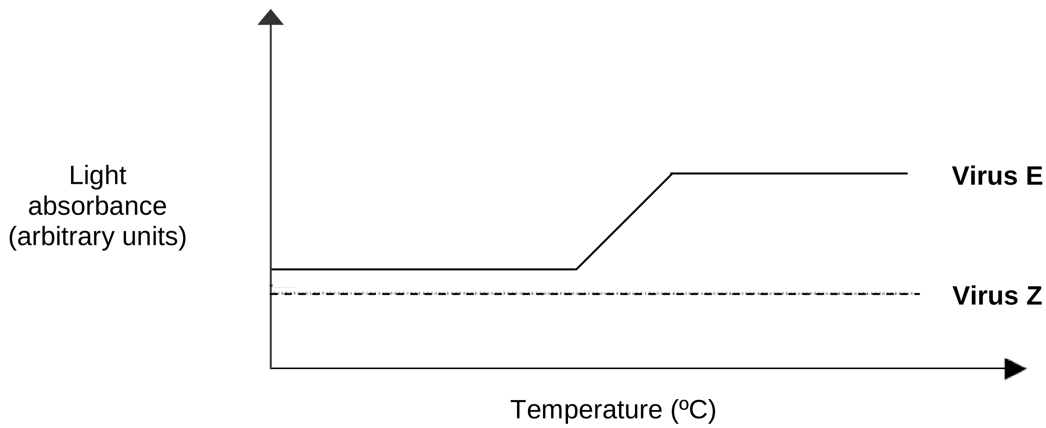
B production of host enzymes by virus which nicks its own DNA such that it can be assembled into the new virus.

C no inhibition of virus DNA production such that the virus DNA can be assembled into the new virus.

D integration of virus DNA into host DNA and during excision, the viral genome carries along with it the host DNA to be assembled into the new virus.

24 Two new **viruses E** and **Z**, which infect eukaryotic cells, have been identified.

In one experiment, the nucleic acid from each virus is isolated and analyzed over a range of temperatures. The light absorbance of nucleic acids changes when denaturation or annealing occurs. The behaviour of the nucleic acid from each virus is shown in the graph.



In a second experiment, it is found that treatment with reverse transcriptase inhibitors or with inhibitors of DNA synthesis blocks the ability of virus Z to infect cells. In contrast, reverse transcriptase inhibitors have no effect on the ability of virus E to infect cells but DNA synthesis inhibitors block infection by virus E.

Which of the following conclusions can be drawn from both the experiments?

A The genome of virus E is single-stranded RNA and that of virus Z is double-stranded DNA.

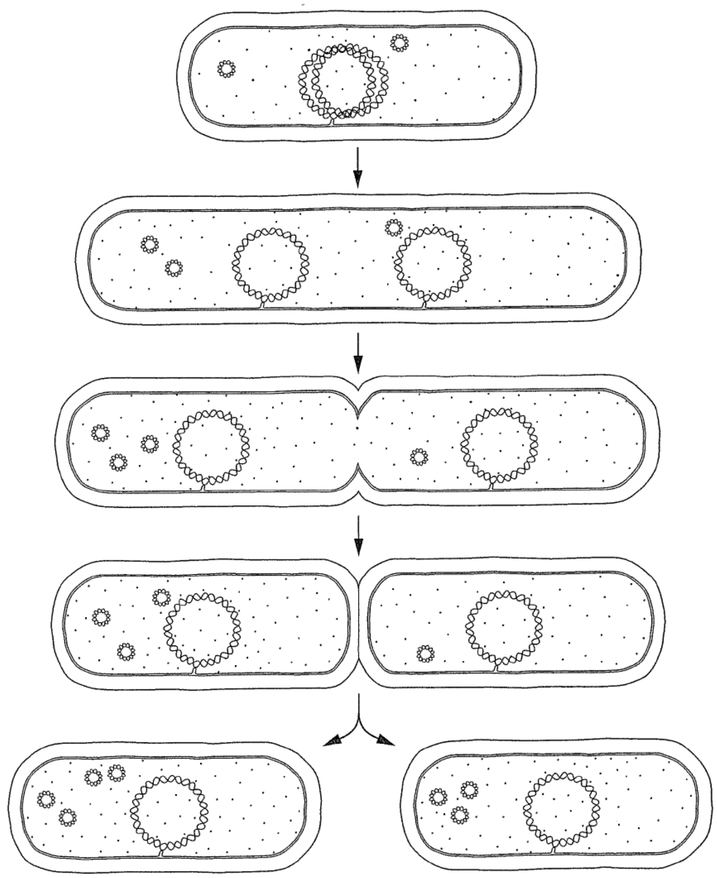
B The genome of virus E is double-stranded DNA and that of virus Z is single-stranded RNA.

C The genome of virus E is double-stranded RNA and that of virus Z is single-stranded DNA.

D The genome of virus E is double-stranded DNA and that of virus Z is double-stranded RNA.

[Turn over

25 The figure shows the reproduction of a bacterial cell.



Which of the following can be concluded from the figure?

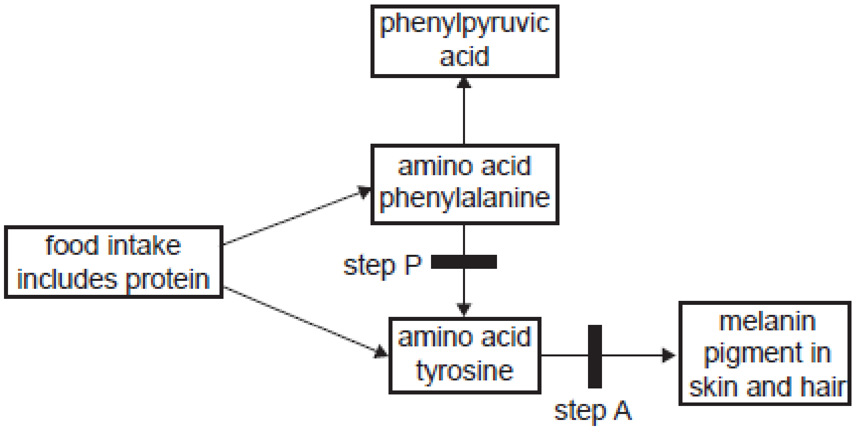
A Some DNA are able to control their own replication.

B This type of reproduction results in the formation of genetically identical offspring.

C Some DNA can be excised to form smaller DNA molecules.

D This type of reproduction divides the cell into two daughter cells by forming a cell plate.

26 The figure below shows part of a metabolic pathway involving the amino acids phenylalanine and tyrosine. One or more enzymes are involved at each step in such a pathway.



Two steps have been highlighted, step **P** and step **A**. Each is under the control of a single gene.

Step **P** is controlled by **gene P** which has the alleles

P : production of phenylalanine hydroxylase

P' : no enzyme

Step **A** is controlled by **gene A** which has the alleles

A : production of tyrosinase

a : no enzyme

A couple, each heterozygous at the **P** and **A** loci, have a daughter Emily.

With reference to these two genes, what is the chance that Emily has the same genotype as her parents?

A 0% **B** 25% **C** 75% **D** 100%

27 The fur colour of rabbits is controlled by a gene with 3 alleles. The phenotypes are black, brown and white fur. 4 crosses were repeated many times. The crosses and the outcomes of these crosses are shown in the table below.

Cross	**Parents**	**Offspring phenotype and ratio**
1	black x black	3 black : 1 white
2	brown x white	1 brown : 1 white
3	black x black	3 black : 1 brown
4	white x white	all white

From the data, it is possible to conclude that

A brown fur is recessive to white fur.

B all of the white fur offspring are heterozygous.

C two thirds of the black fur offspring in cross **3** are heterozygous.

D the black fur parents in cross **1** have the same genotype as the black fur parents in cross **3**.

28 A yellow flower, green-stemmed plant with the genotype **YYrr** was crossed with a white flower, red-stemmed plant with the genotype **yyRR**. The F_1 plants were allowed to self fertilise, A χ^2 test was carried out on the results obtained for the F_2 generation. Part of the table of values for χ^2 is shown.

degrees of freedom	p = 0.5	p = 0.1	p = 0.05	p = 0.01	p = 0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.61	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.37	7.78	9.49	13.28	18.46
5	4.35	9.24	11.07	15.09	20.52

The value of χ^2 in this investigation was 10.6.

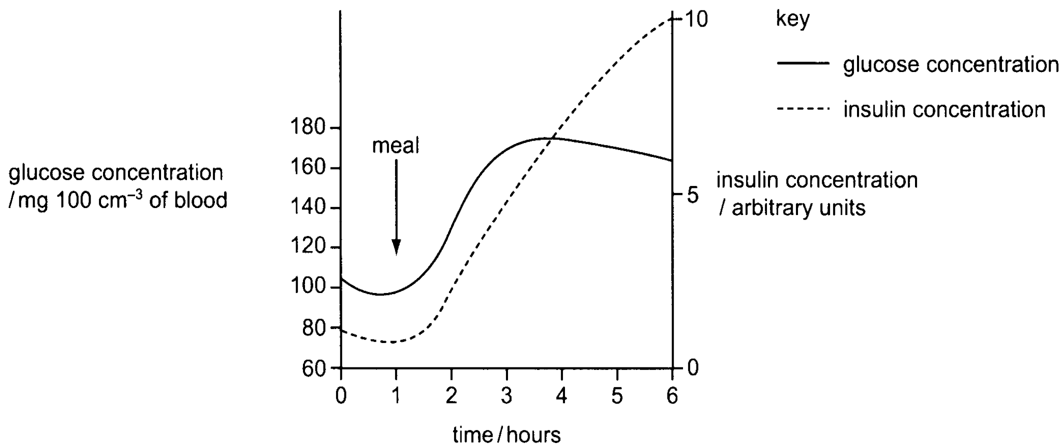
What is the probability of this value and do the results fit the expected ratio?

	Probability	Results fit expected ratio
A	Between 0.01 and 0.05	No
B	Between 0.01 and 0.05	Yes
C	Between 0.05 and 0.1	Yes
D	Between 0.1 and 0.5	No

29 Which combination of definitions is correct?

	Taxonomic group	Phylogeny	Binomial
A	grouping organisms according to how recently they evolved from a common ancestor	survival of the fittest	the names of the genus and species of the organism
B	a category in a classification system	grouping organisms according to how recently they evolved from a common ancestor	the names of the family and genus of the organism
C	a category in a classification system	grouping organisms according to how recently they evolved from a common ancestor	the names of the genus and species of the organism
D	grouping organisms according to how recently they evolved from a common ancestor	survival of the fittest	the names of the genus and species of the organism

30 The graph shows a patient with diabetes who consumes a meal rich in carbohydrates. The concentrations of glucose and insulin in the blood stream were monitored for five hours after the meal. The glucose concentration is measured in mg 100 cm^{-3} and the insulin concentration in arbitrary units.



What would be an effective treatment for this patient?

A Injection of insulin before a meal

B Reduction of dietary carbohydrate

C Use of a glucagon inhibitor

D Use of a diuretic to increase urine production

31 Which of the following is/are true about G-protein signaling?

I Testosterone hormone binds to its cell surface receptor to cause GTP to displace GDP, thus activating G-protein.

II Upon activation of G-protein, the G-protein-linked receptor dissociates to activate adenylyl cyclase.

III GTPase terminates the activity of G-protein by hydrolysis of GTP.

IV The ratio of G-protein-linked receptor to G-protein is 1:1.

A III only

B II and III only

C III and IV only

D I, III and IV only

32 Which of the following statements show the advantages of signal transduction pathways?

1. Amplification of signals.
2. Fine-tuning of cellular responses.
3. Different responses from a single cell.
4. Switching on of certain genes while other genes are switched off.

A 1 only

B 1 and 2

C 2, 3 and 4

D 1, 2, 3 and 4

33 Four events in the transmission of nerve impulses across synapses are:

1 depolarisation of the presynaptic membrane
2 propagation of postsynaptic action potential
3 hydrolysis of transmitter substance
4 rupturing of synaptic vesicles

In which sequence do these events occur?

	First	→		Last
A	1	3	2	4
B	1	4	2	3
C	4	1	3	2
D	4	3	1	2

34 Which is a correct description of the role of calcium ions in the neuromuscular system?

A exchanged with sodium ions through co-transport channels at axon surfaces during the reestablishment of a resting potential after an action potential

B moved in by diffusion through gated ion channels in pre-synaptic membranes of neurones causing vesicles to move to pre-synaptic membrane as an impulse arrives

C exchanged with chloride ions at the post-synaptic membrane, in changing membrane potential in inhibitory neurones

D actively pumped out of axons at nodes of Ranvier of myelinated neurones, being the main cause of the potential difference that is maintained during the resting potential

35 Which statement about anaerobic respiration is correct?

A Animals are unable to use lactate for the production of ATP.

B From one molecule of glucose, ethanol and lactate production yield the same amounts of ATP.

C From one molecule of glucose, ethanol production yields more energy than lactate production.

D Yeast is able to respire ethanol for the production of ATP.

36 In light, Mg^{2+} ions move into the stroma. In the dark, the concentration of Mg^{2+} ions in the stroma is low.

In high Mg^{2+} concentrations, CO_2 and Mg^{2+} react with the active site of RuBP carboxylase (rubisco), making a carbamate group ($C0_2NH$) and making the rubisco active. In low Mg^{2+} concentrations the carbamate group dissociates making the rubisco inactive.

Inactive rubisco will bind tightly at its active site with any RuBP present, so that no rubisco catalysis can take place. An ATPase enzyme called rubisco activase can, in high concentrations of Mg^{2+}, release the RuBP from the rubisco, producing a carbamate group and activating the rubisco.

Which statement is supported by these facts?

A Inactivated rubisco can be reactivated by NADPH made in the light-depending reaction.

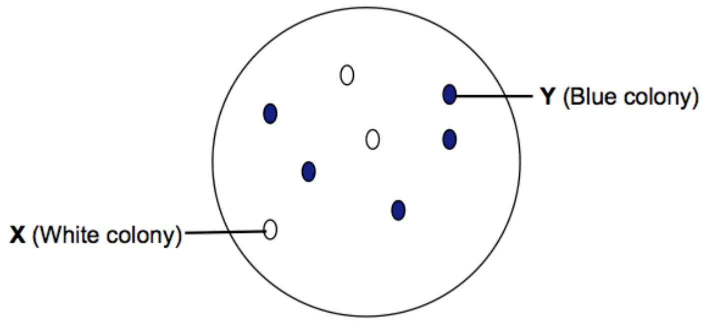
B Low concentrations of 3C compound in the stroma of the chloroplast deactivate rubisco.

C Rubisco without the carbamate group is inhibited by low concentrations of RuBP.

D The rate of carbon dioxide fixation increases when the carbamate group dissociates from rubisco.

37 pGEM-T is a 7.25kb plasmid cloning vector often used in creating recombinant DNA. It contains two selectable marker genes, *TetR* and *lac*Z. The multiple cloning site (MCS) of the cloning vector is located within the *lac*Z gene.

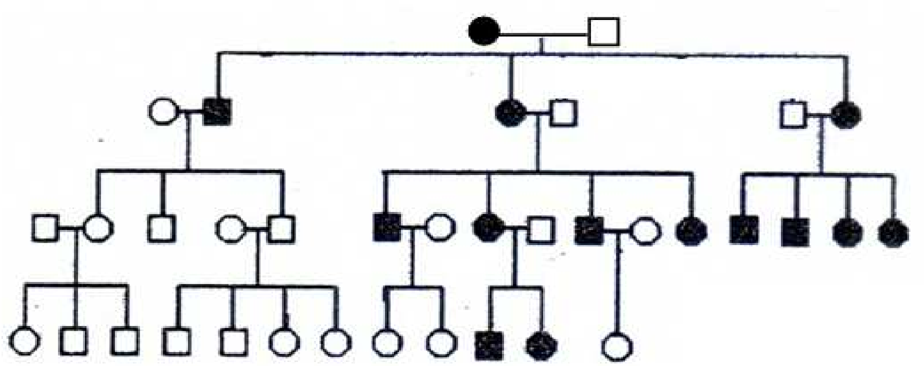
Using *Sac*II as the restriction enzyme, a 2.31kb insert (gene of interest) is cloned into the MCS of plasmid pGEM-T and the recombinant plasmid is transformed into *E. coli* cells. The transformants are then plated on medium containing tetracycline and X-Gal. The following result is obtained.



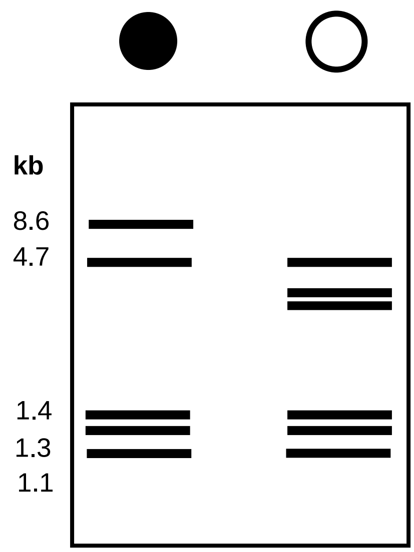
Colonies **X** and **Y** are isolated and the plasmid DNA from the colonies is extracted. The extracted plasmids are then digested with *Sac*II. Which of the following correctly shows the fragments obtained?

	Colony X	Colony Y
A	A 9.56kb fragment	A 7.25kb fragment
B	A 7.25kb fragment	A 9.56kb fragment
C	7.21kb and 0.34kb fragments	7.21kb and 2.31kb fragments
D	7.21kb and 0.34kb fragments	7.21kb and 0.34kb fragments

38 The pedigree chart shows the inheritance of an RFLP marker locus which contains restriction sites for *Hae*II in mitochondrial DNA (mtDNA). mtDNA exists as a double-stranded circular DNA molecule, very much resembling a bacterial plasmid. The filled symbols indicate individuals who have lost a *Hae*II site to a point mutation.



mtDNA from an affected individual and an unaffected individual is digested with *Hae*II to produce restriction fragments that are separated using gel electrophoresis. The results are shown as follows:



Which of the following statements is not correct?

A There were six *Hae*II restriction sites in the mtDNA before the mutation.

B Affected females always transmit the trait to all their children.

C Mitochondrial genes in males are not transmitted to any of their children.

D The mutation occurred at the restriction sites between fragments of 4.1 kb and 4.7 kb.

39 A multiple cloning site

A contains many copies of a cloned gene.

B allows flexibility in the choice of restriction enzymes for cloning.

C allows flexibility in the choice of organism for cloning.

D contains many copies of the same restriction enzyme site.

40 The basis for development of leukemia in some children treated with a retrovirus-derived gene therapy vector is:

A insertion of the viral genome adjacent to an oncogene

B use of a virus that contains an oncogene

C insertion causing a mutation of a tumor suppressor gene

D stimulation of immune response to cell surface antigens introduced by an inserted gene